

LAB-3: Component Modelling & Architectural Pattern Selection

Scenario: EV Charging Station Reservation System

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1. Architectural Style Analysis

Architecture	Advantages	Disadvantages
Layered	Clear separation of concerns; easy to understand and maintain; good for traditional applications	Performance overhead between layers; tight coupling between adjacent layers; harder to scale individual layers
Microservices	Independent scaling; fault isolation; technology diversity; independent deployment	Operational complexity; network latency; data consistency challenges
Client-Server	Centralized control; simple deployment; easy data consistency	Single point of failure; scalability bottleneck; limited fault tolerance

2. Architecture Selection

Selected: Layered Architecture — organized into Presentation, Business, and Data layers.

3. Component Identification

Given: Order Manager Component, Payment Service Component

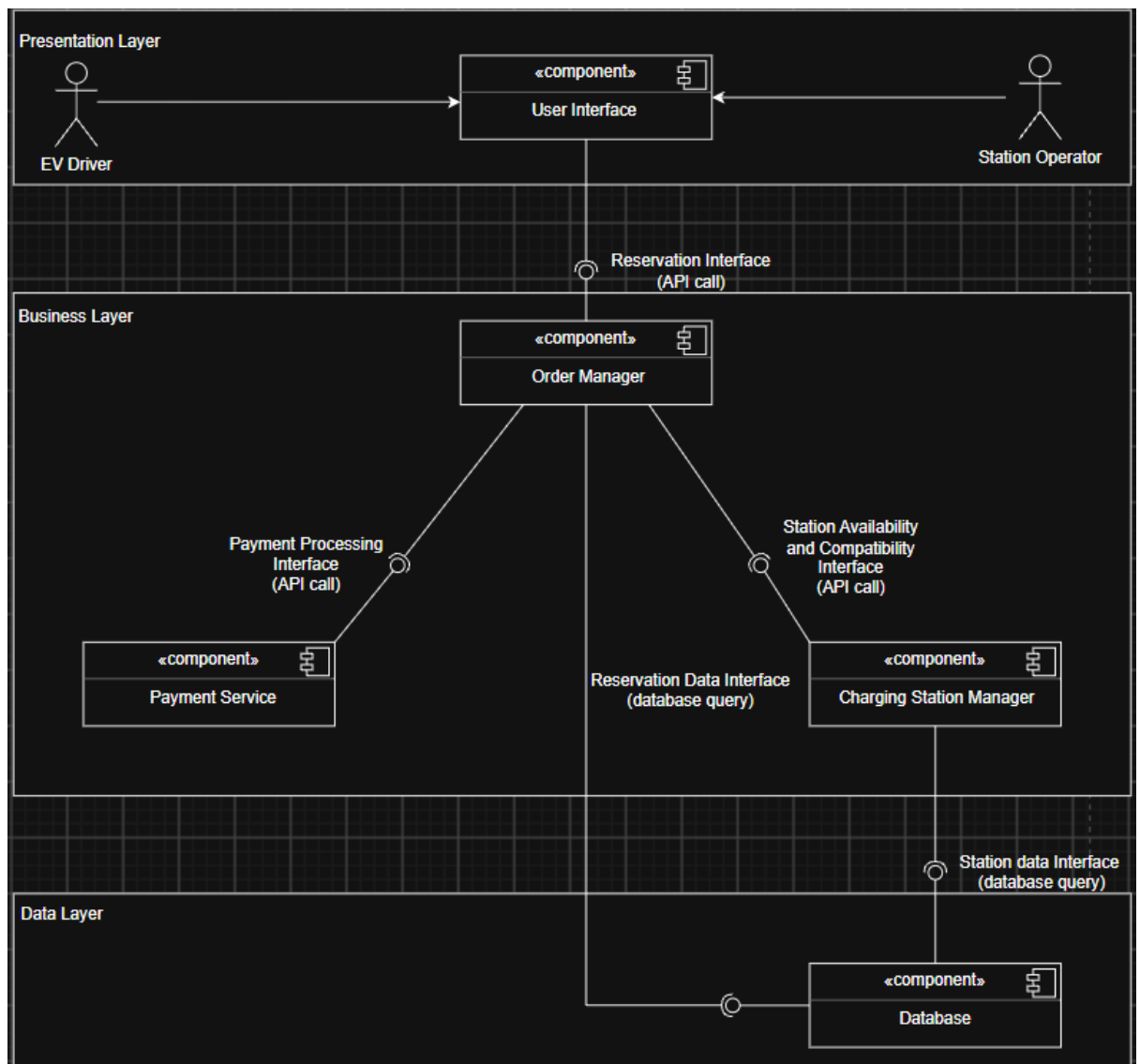
Identified (remaining 3): User Interface Component, Charging Station Manager Component, Database Component

Component	Layer	Responsibility
User Interface Component	Presentation	Driver/operator interaction — station search, connector selection, time-slot selection, reservation, payment initiation
Order Manager Component	Business	Coordinates the reservation workflow — checks availability, triggers payment, records/cancels reservations
Payment Service Component	Business	Processes payment via the external Payment Gateway
Charging Station Manager Component	Business	Manages station/socket availability, connector compatibility, auto-release of expired slots
Database Component	Data	Stores station, socket, connector, reservation, and pricing data

4. Interfaces

#	Interface	Between
1	Reservation Interface	User Interface ↔ Order Manager
2	Payment Processing Interface	Order Manager ↔ Payment Service
3	Station Availability & Compatibility Interface	Order Manager ↔ Charging Station Manager
4	Reservation Data Interface	Order Manager ↔ Database
5	Station Data Interface	Charging Station Manager ↔ Database

5. Component Diagram



6. Written Justification

Architecture Selection

I chose Layered Architecture for the EV Charging Station Reservation System.

Architectural Choice

The system is organized into Presentation, Business, and Data layers. The Presentation Layer contains the User Interface Component. The Business Layer contains the Order Manager, Payment Service, and Charging Station Manager components. The Data Layer contains the Database Component.

Reason 1 — Separation of Concerns

Layered Architecture cleanly separates driver interaction (User Interface), reservation logic (Order Manager), payment processing (Payment Service), and station/connector management (Charging Station Manager) from data storage (Database), making the system easier to build, test and maintain.

Reason 2 — Centralized Reservation Control

The Order Manager acts as a coordination point: it verifies station and connector compatibility with the Charging Station Manager, triggers payment through the Payment Service, and only then commits the reservation to the Database — preventing conflicting or double-booked reservations.

Security Advantage

Payment handling is isolated in the Payment Service rather than exposed to the User Interface directly, and all reservation and payment-related data access is mediated by the Order Manager, reducing direct access to sensitive reservation and payment data.

Performance Benefit

Layering adds some overhead between layers, but the cost is small for this system. Station and socket availability is handled by the dedicated Charging Station Manager rather than by the Presentation Layer, so the most frequent operations (station searches and availability checks) are kept separate from user-interface processing, improving the efficiency of station searches and reservation requests.